Statistical Inference for Policy-Mediated Effects in Bilevel Decision Problems

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Abstract

Bilevel decision problems involve an inner and an outer optimization problem. Bilevel decision problems consist of an inner and an outer optimization problem. The outer variable, denoted by x represents a design parameter, such as a reward/penalty parameter, and the inner variable y represnets a learned response, such as the parameters of a fitted model or decision rule. For a given x, the inner problem minimizes a loss over y, yielding the response $y^\star(x)$. The outer problem then selects an x by minimizing a separate criterion evaluated at x and $y^\star(x)$. In this work, we develop statistical inference for the total derivative of the outer loss with respect to x, separating the direct contribution due to the design parameter from the contribution through the learned response. We construct orthogonal estimators and establish joint asymptotic normality, accounting for dependence within participant trajectories. For a class with affine inner estimating equations and outer losses quadratic in the learned response, we establish matching minimax rates for the response contribution in sample size and lower-level identification strength. We construct a stabilized estimator that attains said rates, we also construct projection confidence sets that retain coverage under weak inner loss identification. We perform a Nonlinear end-to-end simulation to assess finite-sample estimation and inference under varying identification strength and stationary-set geometry. Applications to HeartSteps and Drink Less datasets illustrate how accounting for the learned response affects stationary design decisions and their implied intervention intensities under prescribed benefit–burden criteria.

Keywords: bilevel decision making; semiparametric efficiency; micro-randomized trials; weak identification; outer-decision inference

1 Introduction

Bilevel decision problems have two variables, an outer design choice and an inner response which is linked to that design choice. In it's standard optimization formulation, there is an

inner optimization problem that minimizes a loss to determine a response for a given candidate design parameter. The outer optimization problem then selects the design parameter by minimizing a loss function which is evaluated at both the parameter and the corresponding response. Hyperparameter selection provides a familiar example: an inner problem fits model coefficients for a specified regularization strength, while an outer problem selects that strength by evaluating the resulting model on validation data. This structure is also found in other fields such as in meta-learning, where an outer configuration determines the learning problems solved for individual tasks (Franceschi et al., 2018; Ghadimi and Wang, 2018).

A central statistical question that arises in this setup is how precisely the data determine the *total derivative of the outer loss* with respect to the design parameter. This derivative contains both a direct contribution, due to the variation in the design parameter while holding the response fixed, and a contribution through the change in the learned response. Because the inner fitting loss and the outer evaluation loss can differ, solving the inner problem does not generally eliminate the latter contribution.

We develop statistical inference results for this total derivative and its constituent parts for the case of a scalar design parameter. Our framework allows the inner response to be defined by a smooth, finite-dimensional population estimating equation. We construct confidence sets for stationary design values, at which the total derivative of the outer loss equals zero. Establishing that these values minimize the outer loss requires additional conditions. Section 2 formally defines the optimization problem, its extension to responses defined by estimating equations, and the quantities targeted by our inference procedures.

Adaptive interventions provides a real world use-case of this framework. In sucha a setup, an investigator can choose a penalty on intervention burden as the design parameter. For

each penalty, an inner problem determines a policy governing when support is offered; while an outer criterion evaluates the penalty together with its induced policy under a specified benefit-burden tradeoff. A higher penalty changes the assessed cost of intervention at a fixed policy and can also reduce intervention frequency through the fitted policy response. The fitting criterion may include entropy regularization to produce a smooth policy, while the evaluation criterion excludes that regularization term. In this setting, we call the contribution through the learned response the *policy-mediated component*. This term describes sensitivity through a fitted policy, rather than a natural indirect effect through a post-treatment mediator (Qian, 2025).

We illustrate our methods through retrospective analyses of HeartSteps and Drink Less, two micro-randomized trials. Such trials repeatedly randomize intervention delivery to learn when support is useful, while recognizing that intervention burden matters (Klasnja et al., 2015; Nahum-Shani et al., 2018; Qian et al., 2022). HeartSteps studied suggestions to increase physical activity (Klasnja et al., 2019), and Drink Less studied notifications intended to increase engagement with an alcohol-reduction app (Bell et al., 2023). Our analyses evaluate specified smooth policy responses under benefit–burden criteria; these response rules were not deployed in either trial. Repeated observations within participants require inference that accounts for within-participant dependence, while randomization probabilities known by design provide opportunities to improve statistical efficiency.

Two sources of weak information complicate derivative and decision inference. First, when the data weakly determine the lower response, estimation error in its defining equation can be amplified in the estimated response sensitivity and hence in the total derivative. Second, even when the lower response is well identified, a nearly flat total derivative can leave a

stationary decision poorly localized. Multiple stationary values may also be compatible with the data. These difficulties concern different parts of the problem: identification of the inner response and localization of the outer stationary set. Moreover, evidence for a nonzero contribution through the learned response at one design value need not establish a statistically distinguishable change in a selected stationary decision.

1.1 Related work

The regular estimation problem builds on influence-function theory and orthogonal, cross-fitted estimation (Bickel et al., 1993; Newey, 1994; van der Vaart, 1998; Chernozhukov et al., 2018). The closest work on bilevel-gradient inference is El Khoury et al. (2026), who derive an efficient influence function and a cross-fitted orthogonal estimator for a population bilevel gradient with an infinite-dimensional lower object. They establish asymptotic normality and uniform control of the gradient process. Our analysis treats smooth, finite-dimensional response equations and develops identification-dependent estimation limits, confidence sets under weak lower identification, and simultaneous inference for potentially flat or nonunique stationary decisions. It also accommodates participant trajectories and, when available, known sequential randomization. These frameworks have complementary scopes. Where the targets and statistical models coincide, the efficient influence functions agree (Corollary A.16).

Statistical methods for learning treatment rules and dynamic regimes include value-based policy learning and inference (Murphy, 2003; Qian and Murphy, 2011; Zhao et al., 2012; Zhang et al., 2012; Luckett et al., 2020; Shi et al., 2024). Micro-randomized-trial methods estimate proximal excursion effects (Boruvka et al., 2018; Qian et al., 2021), while longitudinal

policy-inference methods examine long-term average outcomes, value functions, and sequential evaluation (Liao et al., 2021; Shi et al., 2022; Ramprasad et al., 2023). Two recent preprints illustrate related inferential objectives. Wei (2026) develop off-policy inference for optimal policy values under possible nonuniqueness and analyze learned notification policies using Drink Less. Wolf et al. (2026) estimate a stochastic regimen–response curve for distal outcomes, with simultaneous inference for the curve and optimizer inference under additional curvature conditions. Our question concerns sensitivity to a design parameter that induces a learned response, the decomposition of that sensitivity, and its consequences for stationary design decisions.

Our participant-level construction retains dependence among repeated observations within a trajectory (Liang and Zeger, 1986). Known randomization can further restrict the statistical model and lower the efficiency bound, connecting the analysis to covariate augmentation and semiparametrically efficient sequential-policy inference (Zhang et al., 2008; Kallus and Uehara, 2020a,b; Shi et al., 2026).

Weak identification of the lower response connects the problem to inference with weak or nearly singular estimating equations (Stock and Wright, 2000; Andrews and Cheng, 2012; Han and McCloskey, 2019; Kaji, 2021). We quantify estimation limits in a bounded affine–quadratic class using minimax testing arguments (Donoho and Liu, 1991; Tsybakov, 2009). Our confidence sets build on identification-robust testing and projection (Anderson and Rubin, 1949; Stock and Wright, 2000; Dufour and Taamouti, 2005). These constructions can retain coverage when ordinary inference based on an estimated inverse sensitivity matrix is unreliable. Unbounded confidence sets can correctly reflect insufficient identifying information (Dufour, 1997).

Uncertainty about outer stationary values raises a separate nonregular problem, related to nonunique or nonsmooth targets in policy inference (Laber et al., 2014; Luedtke and van der Laan, 2016; Hirano and Porter, 2012; Fang and Santos, 2019). We use Gaussian-process and multiplier-bootstrap approximations (Chernozhukov et al., 2007, 2013, 2014, 2016) to construct and invert a simultaneous band for the total derivative. Under regular lower identification, this approach covers the entire stationary set without requiring prior selection of a single root.

1.2 Approach and contributions

We develop joint statistical inference for the direct contribution, the contribution through the learned response, and the total derivative of the outer loss. Our estimators use orthogonal scores to reduce first-order sensitivity to errors in estimating auxiliary quantities. Cross-fitting and covariance estimation use complete participant trajectories, accounting for within-participant dependence. The analysis yields four principal results.

1. **Joint inference and sequential efficiency.** Under smoothness, regular lower identification, and suitable nuisance-estimation conditions, we establish joint asymptotic normality for the two derivative components and the total derivative at a fixed design value. We derive efficient influence functions in the unrestricted participant-trajectory model. We also characterize the additional efficiency available under known sequential randomization and give an augmentation attaining the smaller bound under its stated conditions.

2. **Identification-dependent estimation accuracy.** For the derivative contribution through the lower response, we establish matching minimax rates in sample size and

identification strength within a bounded, fixed-dimensional affine-quadratic class at a fixed design value. A stabilized estimator attains these rates under specified identification and class bounds. The guarantees quantify how weak lower identification limits attainable estimation accuracy.

3. **Confidence sets under weak identification of the inner response.** We construct confidence sets for the derivative contribution through the learned response at a fixed design value. Under stated moment conditions, joint tests of the response and auxiliary equations yield asymptotically valid coverage without inverting the estimated inner Jacobian or requiring a positive lower bound on identification strength. The sets may be conservative or unbounded. At exact singularity, coverage applies to each selected value compatible with the moment equations, rather than all such values simultaneously.

4. **Inference for stationary decisions.** Under regular lower identification and uniform-process conditions, inversion of a simultaneous band for the total derivative covers the entire stationary set, including flat or nonunique roots. Conventional inference for a selected stationary point follows under additional local uniqueness and nonzero-slope conditions.

The weak-identification and stationary-set guarantees address separate information regimes. In particular, the stationary-set result retains regular identification of the lower response.

A nonlinear end-to-end simulation evaluates the complete numerical estimation and inference procedure, while controlled experiments examine weak inner identification and flat or nonunique outer stationary sets. Applications to HeartSteps and Drink Less illustrate

how the framework connects derivative inference to comparisons of stationary penalties and intervention intensities under specified benefit–burden criteria. The analyses compare decisions obtained with and without accounting for the learned response and quantify uncertainty in the resulting contrasts.

Section 2 defines the complete framework and estimands; Section 3 presents the estimators; and Section 4 states the guarantees. Sections 5 and 6 report the simulations and applications. Proofs, extensions, and additional numerical investigations appear in the supplement.

2 Statistical framework and estimands

We first define the inner and outer optimization problems and then extend the lower response to a population estimating equation. We specify the observed-data models, the derivative components, and the stationary-decision target in this section. Estimation and inference are developed in Sections 3 and 4.

2.1 Inner and outer decision problems

Let $O \sim P$ denote an observational unit under an unknown distribution P. The outer variable $x \in \mathcal{X} \subset \mathbb{R}$ is a scalar design parameter, and the inner variable $y \in \mathbb{R}^p$ is a finite-dimensional response parameter. Here $\mathcal{X}$ is an open interval, and inference for decisions is restricted to a prespecified compact interval $\mathcal{X}_0 \subset \mathcal{X}$. The response may parameterize a fitted predictive model, a policy, or another decision rule.

For lower and outer loss functions ℓ_{in} and ℓ, define the population criteria

$$\begin{aligned} g_P(x, y) &= \mathbb{E}_P\{\ell_{\text{in}}(O; x, y)\}, \\ f_P(x, y) &= \mathbb{E}_P\{\ell(O; x, y)\}. \end{aligned} \tag{1}$$

The standard bilevel optimization formulation is

$$\begin{aligned} y_P^\star(x) &\in \arg\min_{y \in \mathbb{R}^p} g_P(x, y), \\ \Phi_P(x) &= f_P\{x, y_P^\star(x)\}, \qquad \min_{x \in \mathcal{X}_0} \Phi_P(x). \end{aligned} \tag{2}$$

For each candidate x, the inner problem determines a response $y_P^\star(x)$. The outer problem compares candidate values of x together with their corresponding responses. Thus the outer problem does not choose x and y independently.

The inner fitting loss and outer evaluation loss may differ. For example, the inner loss may include a regularization term that is absent from the outer criterion. The outer criterion may also depend explicitly on x, in addition to its dependence through $y_P^\star(x)$. Both forms of dependence enter its total derivative.

These are population targets. A fitted response $\widehat{y}(x)$ estimates $y_P^\star(x)$ from observed data; the population response is defined separately from the numerical algorithm used to estimate it. Throughout, differentiation with respect to x holds the observed-data law P fixed.

We use minimization to introduce the optimization structure. The same derivative analysis applies to a criterion expressed as a benefit to be maximized. Minimizing the negative benefit gives an equivalent optimization problem: negation reverses all derivative components but preserves their stationary values. The intervention example below and the applications in

Section 6 retain the benefit convention.

2.2 Observed data and statistical models

We observe independent units $O_1, \ldots, O_n \overset{\text{iid}}{\sim} P$. For the longitudinal setting studied here, an observational unit is a complete participant trajectory:

$$O_i = (S_{i0}, A_{i0}, R_{i1}, S_{i1}, \ldots, A_{i,T_i-1}, R_{iT_i}, S_{iT_i}), \qquad i = 1, \ldots, n. \tag{3}$$

Here S_{it} is the state or context before action A_{it}, $R_{i,t+1}$ is the subsequent reward or outcome, and T_i is participant i's observed follow-up length, which may vary across participants. For the sequential efficiency results, we assume $T_i \leq H$ for a fixed finite horizon H and pad shorter trajectories with absorbing observations to obtain a common length-H representation.

Observations within a participant may depend on earlier contexts, actions, and outcomes, subject to the regularity conditions used below. Thus n counts independent participants, not decision times. Moment functions and covariance calculations use complete participant records.

We distinguish two statistical models for P.

Baseline model. The participant-trajectory law is unrestricted apart from the stated regularity conditions. Neither a known action-assignment mechanism nor a Markov assumption is imposed. This is the baseline model for the influence functions and efficiency bound.

Known-randomization model. Let H_{it} denote the recorded history immediately before A_{it}, including the current context and past actions and outcomes. The trial protocol specifies

$$A_{it} \mid H_{it} \sim \pi_b(\cdot \mid H_{it}). \tag{4}$$

The assignment rule π_b is known from the trial protocol and governs action assignment in the observed data. This data-collection rule remains fixed as the design parameter x varies. It is distinct from any candidate policy defined by the inner response $y_P^\star(x)$, which may change with x. Apart from the known assignment rule, the initial distribution and history-dependent post-assignment laws remain unrestricted. Randomization probabilities are positive on the actions available under the protocol; when intervention is unavailable, assignment may be deterministically no intervention. Section 3.3 uses knowledge of π_b to improve estimation precision.

For the corresponding factorizations, $S_{i,t+1}$ includes all random post-assignment observations, including the separately displayed reward. An optional Markov restriction replaces the history-dependent next-observation law by $p_t(s_{t+1} \mid s_t, a_t)$; a time-homogeneous version uses one kernel. Appendix E.1 examines the additional efficiency available under these restrictions.

2.3 General lower response and identification

We now generalize the definition of the inner response by allowing it to be specified through a population estimating equation. Let $m(O; x, y) \in \mathbb{R}^p$ collect p estimating functions computed from one observational unit O. For longitudinal data, O is a complete participant trajectory.

We define the inner response by requiring these functions to have population mean zero.

$$M_P(x, y) = \mathbb{E}_P\{m(O; x, y)\}, \qquad M_P\{x, y_P^\star(x)\} = 0. \tag{5}$$

For participant data, m is a function of the complete trajectory.

A differentiable interior minimizer of the lower loss satisfies its first-order condition. When differentiation can be interchanged with expectation, taking

$$m(O; x, y) = \nabla_y \ell_{\text{in}}(O; x, y)$$

gives $M_P(x, y) = \nabla_y g_P(x, y)$. The general formulation also permits moment equations that are not gradients of a lower loss, including suitable actor equations and linear Bellman estimating equations. Consequently, the general results do not require every lower response to be a minimizer or a lower loss to exist.

If the moment equation has multiple roots, we select a response at a reference design value and follow its locally unique continuation where regular identification holds. Results over a design interval require the selected response branch to exist throughout that interval. For an optimization-defined response, a switch between competing minimizers need not produce such a smooth branch; the differentiability conditions must still be verified.

Suppressing P when unambiguous, define

$$A(x) = \partial_y M\{x, y^\star(x)\}, \qquad B(x) = \partial_x M\{x, y^\star(x)\}, \qquad \kappa(x) = \sigma_{\min}\{A(x)\}. \tag{6}$$

The matrix $A(x)$ describes variation of the response equation with its response argument,

while $B(x)$ describes variation with the design parameter at a fixed response. For a gradient equation, these are the lower Hessian and the corresponding mixed derivative. For a general estimating equation, $A(x)$ need not be symmetric.

For the specified moment representation, $\kappa(x)$ measures local identification strength. A small value permits errors in the estimated equation to produce relatively large changes in its solution. The regular results require nonsingularity at the evaluation point, or a uniform positive bound over the relevant interval. The weak-identification results impose different conditions, stated in Section 4.3.

For either an optimization-defined or equation-defined response, the nested criterion remains

$$\Phi_P(x) = f_P\{x, y_P^\star(x)\}, \qquad f_P(x, y) = \mathbb{E}_P\{\ell(O; x, y)\}.$$

Thus the generalization changes how the response is characterized, while retaining the same outer evaluation structure.

Scope of the framework. The general results concern smooth, finite-dimensional population response equations and their outer criteria. Known randomization and Markov restrictions enter only the results stated for those statistical models. All population quantities must have the specified observed-data representations. In particular, if a lower or outer quantity represents a return under a candidate policy, that representation must be identified from P and satisfy the conditions of the applicable inference theorem.

2.4 Derivative components and decision targets

Where the selected response is differentiable, the total derivative of the outer criterion is

$$
\begin{aligned}
D(x) &= \partial_x f\{x, y^\star(x)\}, \\
C(x) &= \left\{ \frac{dy^\star(x)}{dx} \right\}^\top \nabla_y f\{x, y^\star(x)\}, \\
T(x) &= D(x) + C(x) = \frac{d}{dx}\Phi(x).
\end{aligned} \tag{7}
$$

The direct component $D(x)$ varies x while holding the response argument fixed. The component $C(x)$ propagates the change in the response through the outer criterion. We refer to $C(x)$ as the chain component and, when the response parameterizes a policy, as the policy-mediated component.

The lower first-order condition concerns the fitting loss, whereas $C(x)$ involves the response gradient of the outer criterion. Lower-level optimality therefore does not generally make $C(x)$ vanish. If the outer criterion has no explicit dependence on x, the direct component is zero.

For an observational unit, define

$$
d(O; x, y) = \partial_x \ell(O; x, y), \qquad s(O; x, y) = \nabla_y \ell(O; x, y),
$$

with population means

$$
D(x, y) = \mathbb{E}\{d(O; x, y)\}, \qquad \nu(x, y) = \mathbb{E}\{s(O; x, y)\}.
$$

Under the stated smoothness conditions, these are the corresponding partial derivatives of f.

The notation $D(x)$ abbreviates $D\{x, y^\star(x)\}$.

When $A(x)$ is nonsingular, implicit differentiation of (5) gives

$$\frac{dy^\star(x)}{dx} = -A(x)^{-1}B(x), \qquad C(x) = -B(x)^\top A(x)^{-\top}\nu\{x, y^\star(x)\}. \tag{8}$$

This representation connects sensitivity of the response equation to the derivative of the outer criterion.

For a small additive perturbation ΔM of the response equation at a fixed design value, the first-order response perturbation satisfies

$$\Delta y \approx -A(x)^{-1}\Delta M, \qquad \|A(x)^{-1}\Delta M\| \leq \frac{\|\Delta M\|}{\kappa(x)}. \tag{9}$$

Thus weak lower identification can amplify estimation error in the response and its contribution to the outer derivative.

Our outer decision target is the stationary set

$$\mathcal{R}(P) = \{x \in \mathcal{X}_0 : T_P(x) = 0\}. \tag{10}$$

This set may contain one point, multiple points, or an interval. A stationary value need not minimize the nested criterion; additional curvature or global comparisons are needed for that interpretation. Conversely, a boundary minimizer of the constrained design problem need not be a zero of the derivative. The decision guarantees below concern the stationary set.

A zero of D provides a direct-only comparison and is not generally a stationary point of the nested criterion. Detecting a nonzero $C(x)$ at one design value also need not establish

a distinguishable change between selected direct-only and total stationary values. Lower identification and outer-root geometry play different roles: $\kappa(x)$ governs sensitivity of the response equation, whereas the slope and zeros of T govern localization of stationary decisions.

2.5 An intervention-probability response

This example specifies the inner response and outer benefit–burden criterion used in the HeartSteps and Drink Less applications in Section 6. Those analyses use an analytic policy response after estimating a benefit model. The nonlinear end-to-end study in Section 5 uses a related entropy-regularized inner objective, with an additional coefficient penalty and numerically fitted policy parameters.

The observational unit O remains a complete participant trajectory. Let Z denote pre-decision context features extracted from that trajectory, with realization z. The outer design variable x is a penalty per intervention. We first determine the intervention probability at a fixed context and then express the resulting policy through the finite-dimensional response used in the general framework.

Let $b(z)$ denote the population proximal benefit of a binary intervention at context z. For an intervention probability $u \in (0, 1)$ and a fixed $\tau > 0$, define the conditional inner loss

$$\ell_{\text{in}}^{\text{cond}}(z; x, u) = -[b(z) - x]u - \tau h(u), \qquad h(u) = -u \log u - (1 - u) \log(1 - u).$$

Here $b(z)$ already represents an expected benefit conditional on context. Averaging this conditional loss over contexts gives the associated population policy objective. Since intervention

probabilities can be chosen separately at each context, its minimizer is obtained pointwise:

$$\pi_x(z) = \arg\min_{0<u<1} \ell_{\text{in}}^{\text{cond}}(z; x, u) = \text{expit}\{[b(z) - x]/\tau\}, \tag{11}$$

where $\text{expit}(v) = (1 + e^{-v})^{-1}$. Equivalently, the response maximizes entropy-regularized net benefit. The entropy term makes the context-specific solution unique and smooth. Increasing x reduces intervention probability, while smaller τ produces a sharper transition around $b(z) = x$. This candidate policy is distinct from the trial assignment rule π_b that generated the observed data.

To obtain a finite-dimensional representation, suppose $b(z) = z^\top \beta_b$, with an intercept in z. Let e_0 select the intercept, so that $z^\top e_0 = 1$. Then

$$\theta(x) = \frac{\beta_b - x e_0}{\tau}, \qquad \pi_x(z) = \text{expit}\{z^\top \theta(x)\}.$$

Thus u is an intervention probability at one context, whereas θ parameterizes the policy across contexts.

The benefit coefficients are unknown and must be estimated. Suppose they satisfy the population equation

$$M_b(\beta_b) = \mathbb{E}_P\{m_b(O; \beta_b)\} = 0.$$

Define the augmented inner response $y = (\beta_b, \theta)$ and the participant-level estimating function

$$m(O; x, y) = \begin{pmatrix} m_b(O; \beta_b) \\ \\ \theta - (\beta_b - x e_0)/\tau \end{pmatrix}.$$

Its population expectation gives

$$M(x,y) = \mathbb{E}_P\{m(O;x,y)\} = \begin{pmatrix} M_b(\beta_b) \\ \theta - (\beta_b - xe_0)/\tau \end{pmatrix} = 0. \tag{12}$$

This is the response equation in Section 2.3: the first block identifies the benefit coefficients, and the second defines the policy induced by the penalty. Including both coordinates incorporates uncertainty in the learned benefit into inference for the policy response. We use β_b to distinguish the benefit coefficients from the adjoint β introduced in Section 3. Along an identified branch, with P held fixed, $\beta_b'(x) = 0$ and $\theta'(x) = -e_0/\tau$.

For a specified reference distribution of contexts, define the outer benefit criterion

$$f_{\text{ref}}\{x,(\beta_b,\theta)\} = \mathbb{E}_{\text{ref}}\left\{(Z^\top\beta_b - x)\,\text{expit}(Z^\top\theta)\right\} - \frac{\lambda}{2}(x - x_{\text{ref}})^2, \tag{13}$$

where $\lambda > 0$ and x_{ref} express a preference for penalties near a reference value. The outer problem maximizes this criterion after substituting the response from (12). Entropy regularization enters the inner policy construction but is excluded from the outer evaluation criterion.

The reference average $\mathbb{E}_{\text{ref}}$ is distinct from expectation over participant trajectories. In the application intervals, recorded reference contexts are held fixed. For a fixed reference measure, the general loss-minimization formulation is recovered by taking

$$\ell(O;x,y) = -f_{\text{ref}}(x,y), \qquad f_P(x,y) = \mathbb{E}_P\{\ell(O;x,y)\} = -f_{\text{ref}}(x,y).$$

The outer integrand is therefore constant in the participant record; its unknown population dependence enters through the response coordinates. Section 6.1 gives the benefit models, design choices, and uncertainty calculations.

Retaining the benefit-maximization convention, the policy sensitivity is

$$\frac{d\pi_x(z)}{dx} = -\frac{1}{\tau}\pi_x(z)\{1 - \pi_x(z)\}.$$

The direct, chain, and total derivative components are consequently

$$\begin{aligned} D(x) &= -\mathbb{E}_{\text{ref}}\{\pi_x(Z)\} - \lambda(x - x_{\text{ref}}), \\ C(x) &= -\frac{1}{\tau}\mathbb{E}_{\text{ref}}\{[b(Z) - x]\pi_x(Z)[1 - \pi_x(Z)]\}, \\ T(x) &= D(x) + C(x). \end{aligned}$$

The direct component changes the assessed penalty while holding the response fixed; the chain component accounts for the induced change in intervention probability. Negating the outer benefit reverses all three derivative components and preserves their stationary values.

3 Orthogonal estimation and implementation

The targets are the total derivative of the outer criterion and its direct and chain components, as defined in Section 2. Under regular lower identification, the implicit-derivative representation in (8) provides a natural plug-in construction: estimate the lower response, its Jacobian, and the derivatives of the outer criterion, then substitute them into the chain term. Estimation errors in these quantities can introduce first-order bias. We construct orthogonal

scores that remove this first-order sensitivity. Each score is evaluated on an entire participant trajectory, using nuisance quantities fitted on other participants. The procedures in this section use regular lower identification; Section 4.3 instead constructs confidence sets from moment restrictions when lower identification is weak.

3.1 Adjoint representation and orthogonal scores

Fix a design value x for which the lower response is regularly identified, and abbreviate $y^\star = y^\star(x)$. With x held fixed in the following notation, define the participant-level derivative quantities

$$J(O;y) = \partial_y m(O;x,y), \qquad q(O;y) = \partial_x m(O;x,y), \qquad s(O;y) = \nabla_y \ell(O;x,y),$$

whose population means at $y^\star$ are A, B, and ν, respectively. The purpose of the score construction is to preserve the mean of each derivative component while removing its first-order sensitivity to the fitted auxiliary quantities.

In order to do that, we introduce the adjoint equations

$$A\beta = B, \qquad A^\top \alpha = \nu. \tag{14}$$

The solutions are $\beta = A^{-1}B = -dy^\star/dx$ and $\alpha = A^{-\top}\nu$. Thus the chain component can be written in either of two equivalent forms,

$$C = -\beta^\top \nu = -\alpha^\top B.$$

The first expression suggests the participant contribution $-\beta^\top s(O;y)$, but its mean is sensitive to errors in the estimated adjoint β. We therefore add a weighted residual of the equation $A\beta = B$:

$$\psi_{C,0}(O;y,\alpha,\beta) = -\beta^\top s(O;y) + \alpha^\top\{J(O;y)\beta - q(O;y)\}.$$

At the population response and adjoints, the residual has mean zero, so this correction preserves the target mean. At $y = y^\star$, the derivatives of the score mean with respect to the adjoints are

$$\nabla_\alpha \mathbb{E}\psi_{C,0} = A\beta - B, \qquad \nabla_\beta \mathbb{E}\psi_{C,0} = A^\top\alpha - \nu.$$

Both derivatives vanish when (14) holds. Hence, with the response held at its population value, the remaining effect on the score mean is proportional to the product of the two adjoint estimation errors.

Orthogonality with respect to the adjoints is not enough, because the response $y^\star$ must also be estimated. Let a subscript 0 denote a population auxiliary value. Holding α_0 and β_0 fixed, define the remaining response sensitivity by

$$r = \nabla_y \mathbb{E}\{\psi_{C,0}(O;y,\alpha_0,\beta_0)\}|_{y=y^\star}, \qquad A^\top\rho = r.$$

Subtracting $\rho^\top m(O;x,y)$ leaves the target mean unchanged because $\mathbb{E}m(O;x,y^\star) = 0$. Its response derivative is $-A^\top\rho = -r$, cancelling the remaining first-order response sensitivity.

The resulting chain score is

$$\begin{aligned}\psi_C(O;\eta) = & -\beta^\top s(O;y) - \alpha^\top q(O;y) \\ & + \alpha^\top J(O;y)\beta - \rho^\top m(O;x,y),\end{aligned} \tag{15}$$

where $\eta = (y, \alpha, \beta, \rho)$. The derivative with respect to ρ also vanishes at the truth because its moment residual has mean zero.

The direct component requires a similar correction. Although the definition of D differentiates the outer criterion while holding the response fixed, D is evaluated at the unknown $y^\star$. Define

$$r_D = \nabla_y \mathbb{E}\{d(O;x,y)\}|_{y=y^\star}, \qquad A^\top\gamma = r_D.$$

Subtracting a weighted response-equation residual cancels this sensitivity. The direct and total scores are then

$$\psi_D(O;y,\gamma) = d(O;x,y) - \gamma^\top m(O;x,y), \tag{16}$$

$$\psi_T(O) = \psi_D(O) + \psi_C(O). \tag{17}$$

At the population nuisance values, their expectations are D, C, and T, respectively.

At the population auxiliary values, the scores have vanishing first derivatives of their expectations with respect to the nuisance quantities. This Neyman orthogonality eliminates first-order plug-in bias, not all estimation error: products and other higher-order nuisance terms remain (Chernozhukov et al., 2018). Section 3.2 describes participant-level cross-fitting, and Theorem 4.1 states the conditions under which these remainders are negligible for joint

asymptotic inference.

3.2 Cross-fitted estimation and participant-level inference

We now turn the orthogonal scores into estimates of the derivative components and their sum, together with standard errors and pointwise confidence intervals. The procedure fits the inner response and auxiliary quantities on training participants, evaluates scores on held-out participant trajectories, and estimates uncertainty from the resulting participant-level contributions.

Partition the n independent participants into a fixed number $K \geq 2$ of folds $I_1, \ldots, I_K$, with $|I_k|/n \to \lambda_k \in (0,1)$. Keep each participant's full trajectory in one fold: splitting decision times from the same participant across training and evaluation folds would not preserve the intended independence. At each design value x, estimate the nuisances using participants in I_k^c and evaluate their scores on I_k. Write $P_{-k}h = |I_k^c|^{-1} \sum_{i \in I_k^c} h(O_i)$; a subscript $-k$ indicates a training-fold estimate. The following steps give a concrete implementation based on empirical moment equations (fold subscripts on intermediate quantities are suppressed).

1. Estimate the lower response by a consistent local solution $\widehat{y}$ on the selected population branch, satisfying $\|P_{-k} m(O; x, \widehat{y})\| = O_p(n^{-1/2})$. Evaluate the empirical Jacobian and derivatives:

$$\widehat{A} = P_{-k} J(O; \widehat{y}), \qquad \widehat{B} = P_{-k} q(O; \widehat{y}), \qquad \widehat{\nu} = P_{-k} s(O; \widehat{y}).$$

2. Solve the two adjoint systems $\widehat{A}\widehat{\beta} = \widehat{B}$ and $\widehat{A}^\top \widehat{\alpha} = \widehat{\nu}$. Holding these fitted adjoints fixed

when differentiating, estimate

$$\widehat{r} = P_{-k}\nabla_y \psi_{C,0}(O; y, \widehat{\alpha}, \widehat{\beta})\Big|_{y=\widehat{y}}, \qquad \widehat{r}_D = P_{-k}\nabla_y d(O; x, y)|_{y=\widehat{y}}.$$

Solve $\widehat{A}^\top \widehat{\rho} = \widehat{r}$ and $\widehat{A}^\top \widehat{\gamma} = \widehat{r}_D$.

3. For every participant in I_k, evaluate (15) and (16) with the training-fold estimates. Add these held-out components to obtain the total score.

The procedure differentiates the specified moment and loss functions and solves finite-dimensional linear systems; it does not require differentiating through the numerical algorithm used to fit the lower response. Under the stated regularity conditions, Proposition A.11 establishes $O_p(n^{-1/2})$ nuisance rates when the moment and linear-solve residuals have that order. The numerical solution must also track the selected response branch; further implementation conditions are given with that result. Other nuisance estimators may be used when they satisfy Assumption 4.3.

Let $\widehat{\psi}_{F,i}(x)$ be participant i's held-out score for $F \in \{D, C, T\}$. Average the scores across participants, not across decision times:

$$\widehat{F}(x) = \frac{1}{n}\sum_{i=1}^{n} \widehat{\psi}_{F,i}(x), \qquad \widehat{T}(x) = \widehat{D}(x) + \widehat{C}(x). \tag{18}$$

Writing $\widehat{\Psi}_i = (\widehat{\psi}_{D,i}, \widehat{\psi}_{C,i}, \widehat{\psi}_{T,i})^\top$ and $\widehat{\vartheta} = (\widehat{D}, \widehat{C}, \widehat{T})^\top$, estimate the influence-vector covariance by

$$\widehat{\Sigma}(x) = \frac{1}{n}\sum_{i=1}^{n} [\widehat{\Psi}_i(x) - \widehat{\vartheta}(x)][\widehat{\Psi}_i(x) - \widehat{\vartheta}(x)]^\top. \tag{19}$$

The covariance estimate for $\widehat{\vartheta}$ is $\widehat{\Sigma}/n$. Because direct and chain components are computed from the same participant records, their covariance contributes to the variance of the total derivative: $\widehat{\Sigma}_{TT} = \widehat{\Sigma}_{DD} + \widehat{\Sigma}_{CC} + 2\widehat{\Sigma}_{DC}$. Moreover, $T = D + C$ implies that the three-dimensional covariance has rank at most two; joint inference must preserve that identity.

Under Theorem 4.1, at a fixed x and for a component with $\sigma_F(x) > 0$, a pointwise normal interval is

$$\widehat{F}(x) \pm z_{1-\alpha/2}\frac{\widehat{\sigma}_F(x)}{\sqrt{n}}, \qquad \widehat{\sigma}_F^2(x) = \widehat{\Sigma}_{FF}(x), \tag{20}$$

where z_u is the standard normal u-quantile. This interval uses regular lower identification and does not, by itself, provide simultaneous coverage over a design interval. Such coverage requires the process conditions in Section 4.4, as well as numerical optimization and interpolation errors negligible on the inference scale. When the lower Jacobian is weakly identified, Section 4.3 instead uses moment projection without inverting that Jacobian.

3.3 Using known randomization to improve precision

In the known-randomization model of Section 2, the trial assignment probabilities are specified by the protocol. They can be used to reduce score variance through a mean-zero adjustment, without changing the derivative targets. For each component F, define

$$\mu_{F,t}(h, a) = \mathbb{E}\{\psi_F(O; \eta_{F,0}) \mid H_t = h, A_t = a\},$$

where $\eta_{F,0}$ collects the population score nuisances. This is the conditional mean of the entire participant-level score, not merely the immediate outcome at time t. Fit the regressions

$\widehat{\mu}_{F,t,-k}$ on the training participants and, for each held-out participant $i \in I_k$, use

$$\widehat{\psi}_{F,i}^{\text{known}} = \widehat{\psi}_{F,i} - \sum_{t=0}^{H-1} \Big[\widehat{\mu}_{F,t,-k}(H_{it}, A_{it}) - \int \widehat{\mu}_{F,t,-k}(H_{it}, a) \pi_b(da \mid H_{it}) \Big]. \tag{21}$$

Each bracketed contrast subtracts the conditional prediction averaged over the known randomized action distribution from the prediction for the observed action. Conditional on the recorded pre-assignment history and training data, its expectation is zero even if the fitted regression is imperfect. Thus the adjustment preserves the baseline score's mean without requiring a correctly specified regression.

For a binary action with $p(h) = \pi_b(1 \mid h)$ and both actions available, the contrast is

$$(A_t - p(H_t))\{\widehat{\mu}_{F,t,-k}(H_t, 1) - \widehat{\mu}_{F,t,-k}(H_t, 0)\}.$$

Accurate regressions remove the component of score variation associated with randomized assignments. The variance comparison is given in Theorem A.21; attaining the corresponding efficiency bound requires the $L_2(P)$ regression consistency in Theorem A.25.

To preserve the decomposition $T = D + C$, fit the direct and chain regressions and define the total regression as their sum. Then $\widehat{\psi}_{T,i}^{\text{known}} = \widehat{\psi}_{D,i}^{\text{known}} + \widehat{\psi}_{C,i}^{\text{known}}$. Average and center the adjusted scores as in Section 3.2. The efficiency result is pointwise in x. The main uniform-process results concern the baseline scores; applying simultaneous inference to adjusted curves additionally requires uniform consistency of the regressions and empirical-process control of their fitted classes.

Table 1: Settings and guarantees. Each result uses the conditions stated in its subsection. Weak lower identification and outer flatness are treated separately; the stated guarantees do not cover their joint occurrence without additional assumptions.

Procedure	Setting	Guarantee
Orthogonal cross-fitting	Regular lower identification	Efficient pointwise joint inference for D, C, T
Stabilized affine estimator	Fixed x; specified affine–quadratic class bounds	Matching minimax rates for C within that class
Moment projection	Fixed x; standardized moments; no positive identification bound	Coverage of each selected representable value of C
Simultaneous inversion	Regular lower identification; uniform process control	Simultaneous coverage of the entire stationary set

4 Statistical guarantees and confidence sets

The results concern distinct inferential regimes (Table 1). Regular lower identification permits joint pointwise inference for the direct, chain, and total derivatives. The finite-sample minimax and weak-identification results concern the chain component at a fixed design value. Inference for the entire stationary set instead requires uniform control over the design interval and retains regular lower identification. Weak lower identification and flat or nonunique outer roots are therefore treated separately. Proofs appear in Appendices A–D; Appendix E gives efficiency and model-verification results.

4.1 Regular estimation and efficiency

At a fixed design value x, regular inference requires smooth moments, a locally identified response, and negligible orthogonal-score remainders. Participants, rather than their repeated decision times, are the independent sampling units.

Assumption 4.1 (Primitive smoothness and an integrable envelope)**.** *The dimension p is*

fixed. On an open neighborhood U of a population root (x_0, y_0), $m(O; x, y)$ and $\ell(O; x, y)$ are three times continuously differentiable almost surely. Their values and all mixed derivatives through order three are bounded in norm by a common envelope $W(O)$ with $\mathbb{E}W^{2+\delta} < \infty$ for some $\delta > 0$.

Assumption 4.2 (A regular population root)**.** *At the target, $M(x_0, y_0) = 0$ and $A_0 = \partial_y M(x_0, y_0)$ is nonsingular. The response is the locally unique continuation of this root. For results over a compact interval, this branch must exist in the interior of U throughout the interval, with $\inf_x \sigma_{\min}\{A(x)\} > 0$. Global uniqueness and convexity of the lower problem are not required.*

Assumption 4.3 (Accuracy of the fitted nuisances)**.** *For the fixed participant folds in Section 3.2, each fold-trained nuisance converges in probability to its population value and lies in a fixed compact neighborhood of that value with probability tending to one. Write $r_j = \|\widehat{j} - j_0\|$ for $j \in \{y, \alpha, \beta, \rho, \gamma\}$. Uniformly over the folds,*

$$\begin{aligned} r_y^2 + r_\alpha r_\beta + r_y(r_\alpha + r_\beta + r_\rho) &= o_p(n^{-1/2}), \\ r_y^2 + r_y r_\gamma &= o_p(n^{-1/2}). \end{aligned} \tag{22}$$

The two bounds control the chain and direct remainders, respectively. All errors being $o_p(n^{-1/4})$ is sufficient; unequal rates are allowed. Proposition A.11 verifies pointwise $O_p(n^{-1/2})$ nuisance rates for the construction in Section 3.2, with constants that may depend on identification strength.

Theorem 4.1 (Efficient participant-level estimation of direct, chain, and total derivatives)**.** *Under Assumptions 4.1 to 4.3 and iid participant sampling, the scores in (15)–(17) have*

the correct means and are Neyman orthogonal at the truth. Let $\vartheta = (D, C, T)^\top$ *and* $\phi = (\phi_D, \phi_C, \phi_T)^\top$*, where* ϕ_F *is the score at its population nuisance values, centered by* F*. Then*

$$\widehat{\vartheta}(x) - \vartheta(x) = \frac{1}{n}\sum_{i=1}^{n} \phi(O_i; x) + o_p(n^{-1/2}), \qquad \phi_T = \phi_D + \phi_C. \tag{23}$$

Consequently, $\sqrt{n}(\widehat{\vartheta} - \vartheta)$ *converges to* $N(0, \Sigma)$*, where* $\Sigma = \mathbb{E}(\phi\phi^\top)$ *is consistently estimated by* (19)*. The estimated covariance of* $\widehat{\vartheta}$ *is* $\widehat{\Sigma}/n$*; the joint limit may be singular because* $T = D + C$*.*

In the unrestricted participant-trajectory model, the influence functions are canonical gradients and determine the semiparametric efficiency bounds. If the remainder conditions also hold along regular $n^{-1/2}$ *local submodels, the estimators are regular and attain these bounds.*

Known randomization can reduce variance without changing the derivative being estimated. The adjustment in (21) removes the part of the participant score explained by the randomized action, conditional on its history. Its population projection is the efficient influence function in the known-randomization model, and the variance gain is the squared norm of the removed component. The gain is zero exactly when all such action contrasts vanish. Theorems A.21 and A.25 and corollary A.22 give the projection, exact variance comparison, and sufficient regression conditions for efficiency attainment. These statements are pointwise in x; Appendix E gives the further Markov refinements.

4.2 Identification strength and finite-sample accuracy

We next quantify how lower identification affects finite-sample accuracy in a specified affine–quadratic class at $x_0 = 0$. The result concerns the chain component at a fixed design value, not the general nonlinear model. The observed tuple is (J, b, v, H, c), with $\mathsf{H} = \mathsf{H}^\top$, and

$$m(O; x, y) = Jy - b - xv, \qquad s(O; 0, y) = \mathsf{H}y + c.$$

Thus $y^\star(x) = A^{-1}(\mathbb{E}b + x\mathbb{E}v)$, where $A = \mathbb{E}J$. Write $y_0 = y^\star(0)$, $v_0 = \mathbb{E}v$, and $\nu_0 = (\mathbb{E}\mathsf{H})y_0 + \mathbb{E}c$; the target is $C(P) = v_0^\top A^{-\top}\nu_0$.

Fix p and constants $R, L \geq 0$, $V \geq 1$, and $M \geq \sqrt{p+1}$. The class $\mathcal{P}_{\mathrm{aff}}(\kappa)$ in Definition B.5 imposes

$$\sigma_{\min}(A) \geq \kappa, \quad \|y_0\| \leq R, \quad \|v_0\|, \|\nu_0\| \leq V, \quad \|\mathbb{E}\mathsf{H}\|_{\mathrm{op}} \leq L,$$

and second moments bounded by M^2 for each observed primitive, using Frobenius norms for matrices. All constants except κ are fixed independently of n and κ. Here κ is a specified class lower bound; bounds on the response and gradient are additional restrictions, not consequences of affineness.

The stabilized estimator $\widetilde{C}_n$ uses sample means and the specified bounds κ, R, V. It thresholds unstable sample Jacobians, projects response and gradient estimates onto the class bounds, and clips the inverse-moment estimate. The full construction and constants are given in (32) of Appendix B.

Theorem 4.2 (Finite-sample minimax optimality in the affine–quadratic class)**.** *For* $q \in$

$\{1, 2\}$, *define the minimax risk and rate*

$$R_{n,q}(\kappa) = \inf_{\widehat{C}} \sup_{P \in \mathcal{P}_{\text{aff}}(\kappa)} (\mathbb{E}_P |\widehat{C} - C(P)|^q)^{1/q}, \qquad r_n(\kappa) = \min\{\kappa^{-1}, n^{-1/2}\kappa^{-2}\}.$$

For every $n \geq 1$ *and* $0 < \kappa \leq 1/2$,

$$\frac{1}{16} r_n(\kappa) \leq R_{n,q}(\kappa) \leq \sup_{P \in \mathcal{P}_{\text{aff}}(\kappa)} (\mathbb{E}_P |\widetilde{C}_n - C(P)|^q)^{1/q} \leq K_* r_n(\kappa), \tag{24}$$

where $K_* < \infty$ *depends only on the fixed class constants. The infimum is over all measurable estimators based on the* n *participant tuples. The estimator attains the rate for both expected absolute error* ($q = 1$) *and root mean squared error* ($q = 2$).

In the $n^{-1/2}\kappa^{-2}$ regime, halving identification strength requires sixteen times as many participants for the same error rate. The expected-loss sample complexity is $n \asymp \epsilon^{-2}\kappa^{-4}$ for $0 < \epsilon < 1/(16\kappa)$; uniform absolute or mean square consistency along $\kappa_n \downarrow 0$ is possible exactly when $n\kappa_n^4 \to \infty$. This consistency threshold concerns absolute estimation error, not automatic validity of Wald inference (Appendix C). The fixed-x minimax result depends on the stated class and specified bound κ; adaptation to unknown identification strength is not claimed. Appendix B provides the proof, constants, and sample-complexity bounds.

4.3 Confidence sets under weak lower identification

Unlike the stabilized estimator, projection confidence sets require no specified positive identification bound. At fixed x, for each candidate value c of the chain component, we jointly test the response and adjoint equations without inverting the lower Jacobian. Let

$\theta = (y, \alpha, \beta)$ and stack their participant contributions:

$$G(O; \theta, c) = \begin{pmatrix} m(O; x, y) \\ J(O; y)\beta - q(O; y) \\ J(O; y)^\top \alpha - s(O; y) \\ \beta^\top s(O; y) + c \end{pmatrix} \in \mathbb{R}^{3p+1}. \tag{25}$$

The blocks check the response equation, the two adjoint equations, and the target relation $C = -\beta^\top \nu$, respectively. At the true values, all have mean zero when A is nonsingular.

Write $G_i = G(O_i; \theta, c)$, $\bar{G}_n = n^{-1} \sum_i G_i$, and $\widehat{\Sigma}_G = n^{-1} \sum_i (G_i - \bar{G}_n)(G_i - \bar{G}_n)^\top$. The test statistic standardizes their joint discrepancy:

$$Q_n(\theta, c) = \begin{cases} n \bar{G}_n^\top \widehat{\Sigma}_G^\dagger \bar{G}_n, & \bar{G}_n \in \text{range}(\widehat{\Sigma}_G), \\ +\infty, & \text{otherwise.} \end{cases} \tag{26}$$

Here $\dagger$ is the Moore–Penrose inverse of the moment covariance, not of the lower Jacobian; set $Q_n = 0$ if both mean and covariance are zero. The range condition rejects a violated constraint with zero variability instead of discarding it. Coverage requires the following uniform control of the standardized moments, also stated in Assumption C.1.

Assumption 4.4 (Standardized moment-test regularity)**.** *Let $d = 3p + 1$ be fixed and let Θ, possibly unbounded, contain all nuisance representations under consideration. For every $P \in \mathcal{P}_{\text{weak}}$ and target c_P, there is a deterministic $\theta_P \in \Theta$ such that*

$$\mathbb{E}_P G(O; \theta_P, c_P) = 0. \tag{27}$$

At these representing pairs, the iid participant moments satisfy the uniform covariance-whitened $(2+\delta)$-moment bound and the covariance-range and consistency conditions in Assumption C.1, for some fixed $\delta > 0$. Zero-variance constraints are imposed exactly. Appendix C gives the full standardized formulation, including singular covariances.

Theorem 4.3 (Identification-robust projection confidence set)**.** *Under Assumption 4.4, define*

$$\mathcal{C}_{\text{weak}}(1-\alpha) = \{c : \inf_{\theta\in\Theta} Q_n(\theta, c) \le \chi^2_{3p+1,1-\alpha}\}.$$

Take $c_P = C(P)$ when A is nonsingular, or a selected moment-representable value satisfying (27) at singularity. Then

$$\liminf_{n\to\infty} \inf_{P\in\mathcal{P}_{\text{weak}}} \mathbb{P}_P\{c_P \in \mathcal{C}_{\text{weak}}(1-\alpha)\} \ge 1-\alpha.$$

No positive lower bound on $\sigma_{\min}(A)$ is required. At exact singularity the guarantee covers each representable target value, not the entire identified set simultaneously.

Projection retains c if at least one nuisance representation is not rejected. Sets may be unbounded, and the full $(3p+1)$-degree cutoff is conservative when moments are redundant or only c is of interest. Thus the procedure is an identification-robust baseline, not a shortest-length interval. Subvector and calibrated projection methods suggest possible refinements (Andrews, 2017; Kaido et al., 2019), but require separate verification for these moments. The mathematical infimum must include a representing nuisance; bounded numerical searches need justification (Appendix C). The proof is Theorem C.3; Remark B.4 distinguishes absolute-consistency and Wald-inference thresholds.

4.4 Inference for the outer decision

Even under regular lower identification, flat or multiple roots can make a selected stationary point difficult to locate. We therefore retain every design value where a simultaneous band for T contains zero. Covering the derivative curve then covers its entire stationary set without selecting a root first; this requires interval-wide control beyond pointwise cross-fitting.

Assumption 4.5 (Additional control for inference over an interval)**.** *On the compact interval $\mathcal{X}_0$, the response graph lies in an interior compact subset of U and the uniform identification condition in Assumption 4.2 holds. Nuisance consistency and the product bounds in Assumption 4.3 hold uniformly in x. With probability tending to one, fitted nuisance curves lie in a common bounded set and are Lipschitz in x with a fixed constant. The same primitive envelope W applies there, and $0 < \underline{\sigma} \le \sigma_T(x) \le \overline{\sigma} < \infty$ on the studentized region. Multiplier weights are independent standard normals. For exact nominal quantiles, the limiting supremum distribution is continuous at its $(1-\alpha)$ quantile.*

These conditions control nuisance errors and fluctuations across the whole interval. Lemmas D.1 and D.2 derive the process expansion and bootstrap validity; the latter also allows conditional entropy control in place of the common Lipschitz condition. Uniform $L_2(P)$ convergence alone is insufficient.

The band has the form $\widehat{T}(x) \pm \widehat{c}_{1-\alpha}\widehat{\sigma}_T(x)/\sqrt{n}$. To calibrate it, use independent standard normal weights $\xi_1, \ldots, \xi_n$ and define

$$\widehat{\mathbb{G}}_n^{\xi}(x) = \frac{1}{\sqrt{n}\,\widehat{\sigma}_T(x)} \sum_{i=1}^{n} \xi_i \{\widehat{\psi}_{T,i}(x) - \widehat{T}(x)\}. \tag{28}$$

Each weight multiplies one participant's entire score curve. Let $\widehat{c}_{1-\alpha}$ be the conditional $(1-\alpha)$ quantile of $\sup_{x\in\mathcal{X}_0} |\widehat{\mathbb{G}}_n^{\xi}(x)|$.

Theorem 4.4 (Simultaneous confidence sets for flat or nonunique outer decisions)**.** *Under the pointwise conditions of Theorem 4.1 and $\sigma_T(x^\star) > 0$, pointwise studentized inversion using $z_{1-\alpha/2}$ covers a prespecified root $x^\star$ with asymptotic probability $1-\alpha$, without requiring a unique root or a nonzero slope. Under Assumption 4.5 and the conditions of Theorem 4.1, the simultaneously calibrated inverse set*

$$\widehat{\mathcal{X}}_{1-\alpha}^{\mathrm{sim}} = \left\{ x \in \mathcal{X}_0 : \frac{\sqrt{n}\,|\widehat{T}(x)|}{\widehat{\sigma}_T(x)} \le \widehat{c}_{1-\alpha} \right\} \tag{29}$$

satisfies

$$\liminf_{n\to\infty} \mathbb{P}\{\mathcal{R}(P) \subseteq \widehat{\mathcal{X}}_{1-\alpha}^{\mathrm{sim}}\} \ge 1-\alpha,$$

including when T vanishes on a nondegenerate interval.

Pointwise inversion covers one prespecified root; simultaneous inversion covers all roots. Appendix D proves both claims.

For an interior, locally unique root with nonzero slope $H_\star = T'(x^\star)$, Corollary D.4 gives the usual linear expansion. Its standard error is approximately $\sigma_T(x^\star)/(\sqrt{n}\,|H_\star|)$, illustrating loss of precision as the slope vanishes. In the known-randomization model, the efficient root influence function is $-H_\star^{-1}\phi_T^{\mathrm{known}}$ under the corresponding conditions. Uniform inference with augmented curves also needs the control in Section 3.3. Appendix D gives the root expansion, local decision-shift formula, and vanishing-curvature analysis.

5 Simulation studies

Three studies examine the complete numerical procedure, weak lower identification, and outer-root geometry. The latter two isolate regimes that the first study does not reproduce. Throughout, n counts independent participants and nominal confidence levels are 95%. Appendix F gives designs, additional results, and the reproduction map.

5.1 Nonlinear end-to-end estimation and inference

We first estimate the lower response and all four adjoints numerically, rather than supplying true or artificially perturbed nuisances. Each participant contributes 30 randomized decision times with serially correlated activity, accumulated intervention burden, and outcome errors. A nonlinear benefit function depends on activity, burden, and time. The randomized-action pseudo-outcome U identifies that benefit. The inner response has four logistic coefficients, $\pi_y(z) = \mathrm{expit}(z^\top y)$, learned by minimizing

$$L(x, y) = -\mathbb{E}\left[\frac{1}{30}\sum_{t=1}^{30}\{(U_t - x)\pi_y(Z_t) + \tau h(\pi_y(Z_t))\}\right] + \frac{\eta}{2}\|y\|^2,$$

where $h(p) = -p\log p - (1-p)\log(1-p)$, $\tau = 0.5$, and $\eta = 0.05$. The outer benefit omits the entropy and coefficient penalties and includes $-0.7(x - 1.6)^2/2$. We follow a smooth, locally identified lower solution on $x \in [0, 2]$. We retain benefit derivatives in this study; negating the outer criterion gives the minimization convention without changing its stationary points.

Two-fold cross-fitting estimates the response, the adjoints $\alpha, \beta, \rho, \gamma$, and held-out participant scores. The operational solver stops at gradient infinity norm 10^{-8} and relative adjoint

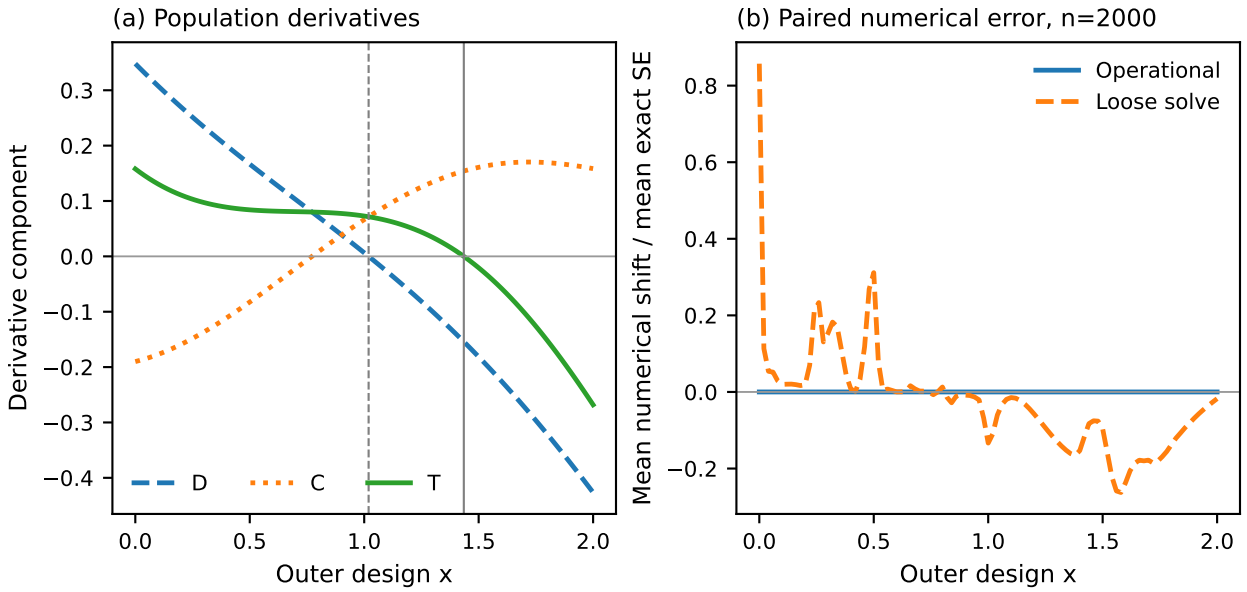


Figure 1: Nonlinear end-to-end study. Left: population direct, chain, and total benefit derivatives; vertical lines mark the direct and total stationary points. Right: mean numerical shift from the tighter fit, divided by its mean standard error, on 200 paired $n = 2,000$ datasets. Operational error is negligible, while the loose solve leaves material error away from the total root.

residual 10^{-8}, using four-dimensional linear solves. Its stopping rule does not use an exact reference response. We compare it afterward with a tighter numerical reference on the same data and folds. The population direct and total stationary points are $x_D^\star = 1.01927$ and $x_T^\star = 1.43517$, a difference of 0.41590. The derivative curves in Figure 1 show that omitting adaptation can reverse a local adjustment direction. Population targets use an independent million-participant calculation, checked by deterministic integration in Appendix F.1.

The primary comparison uses 1,000 independent datasets at each of $n = 500$ and $2,000$, with 1,000 participant multiplier draws and 1,000 paired linearized bootstrap draws per dataset. Bands assess the entire 101-point grid, while root-contrast intervals recompute both selected crossings from resampled participant influence curves. The operational and tighter procedures have identical coverage events; their maximum pointwise T differences at the

Table 2: End-to-end nominal 95% coverage, in percentages; parentheses give Monte Carlo standard errors in percentage points. Operational rows use all 1,000 datasets; loose-solve comparisons use the first 200 paired datasets. Bands concern the 101-point grid; contrasts use paired linearized participant resampling.

n	Method	Rep.	$T(x_T^\star)$	Grid band	$x_T^\star - x_D^\star$
500	Operational	1000	94.8 (0.7)	94.9 (0.7)	95.3 (0.7)
	Loose: orthogonal	200	94.5 (1.6)	92.0 (1.9)	92.5 (1.9)
	Loose: plug-in	200	75.0 (3.1)	0.0 (0.0)	24.5 (3.0)
2000	Operational	1000	94.3 (0.7)	94.2 (0.7)	94.2 (0.7)
	Loose: orthogonal	200	94.5 (1.6)	89.5 (2.2)	92.5 (1.9)
	Loose: plug-in	200	66.5 (3.3)	0.0 (0.0)	14.0 (2.5)

population total root are below 1.4×10^{-12}. Table 2 reports coverage for $T(x_T^\star)$, the grid band, and $x_T^\star - x_D^\star$. The primary coverage estimates have Monte Carlo standard errors of about 0.7 percentage points.

A separate controlled accuracy comparison uses the first 200 paired datasets per size. Gradient ascent stops at distance $n_{\text{train}}^{-0.35}$ from the tighter response; this reference-dependent rule is a stress experiment, not an operational stopping criterion. Orthogonal correction greatly improves pointwise coverage over the nonorthogonal plug-in, but the loose orthogonal band still undercovers: 89.5% at $n = 2,000$. Figure 1 locates numerical error that evaluation at the root alone can miss. The observed remainder therefore matters across the full design interval, even under an asymptotically sufficient error schedule.

In a separate original experiment with 200 paired datasets per size, known-randomization augmentation uses fitted gradient-boosting regressions on whole-participant score targets. At the population total root, variance falls by 40.7% at $n = 500$ and 45.6% at $n = 2,000$. This evaluates actual fitted augmentation rather than supplying a projection by construction; it does not establish attainment of the efficiency bound. Full numerical schedules, fold

sensitivity, dependence checks, and an exploratory full-refit bootstrap comparison appear in Appendix F.1.

5.2 Accuracy and confidence sets as lower identification weakens

The nonlinear experiment retains regular lower identification. To examine weaker identification separately, we first evaluate the stabilized estimator of Theorem 4.2 on scalar and two-dimensional affine–quadratic families at fixed $x = 0$. We use five identification bounds from 0.02 to 0.50, five sample sizes from 100 to 10,000, 13 parameter values per cell, and 5,000 replications per value. The largest grid mean absolute error divided by $r_n(\kappa) = \min\{\kappa^{-1}, n^{-1/2}\kappa^{-2}\}$ ranges from 0.662 to 0.997 in the scalar family and 0.059 to 0.132 in the two-dimensional family (Figure 2, left). These finite-grid maxima illustrate the rate; they do not estimate the supremum over the full model class.

The scalar inverse-moment experiment observes $J_i \sim N(\kappa, 1)$ and targets $C = -1/\kappa$. Figure 2, right, compares inverse-Wald intervals with projection of a confidence interval for κ through $c = -1/\kappa$, using $n = 500$ and 10,000 replications per identification value. Projection retains near-nominal coverage as Wald coverage deteriorates. Along $\kappa_n = n^{-a}$ at $n = 10,000$, projection coverage is 0.9476 and 0.9496 for $a = 0.50$ and 0.60, versus Wald coverage 0.8068 and 0.6221. Projection sets are unbounded in 82.21% and 93.08% of these samples, reflecting uncertainty about magnitude. These are inverse-moment specializations, rather than numerical tests of the general nonlinear moment-projection problem.

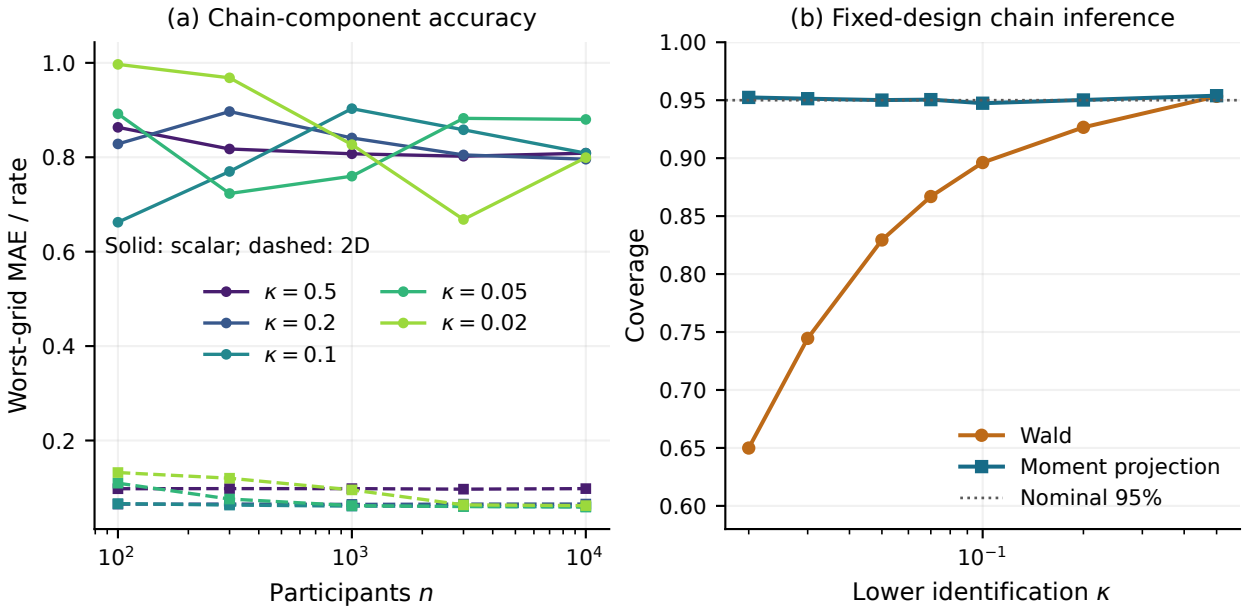


Figure 2: Controlled lower-identification studies. Left: largest grid mean absolute error divided by the minimax rate; solid and dashed lines denote scalar and two-dimensional families. Right: fixed-design chain coverage at $n = 500$, with 10,000 replications per identification value. The accuracy and confidence-set panels use their separately stated models.

5.3 Flat and nonunique outer stationary sets

To isolate outer geometry, we observe $\widehat{T}(x) = T(x) + n^{-1}\sum_i Z_i(x)$, with independent participant curves having centered Gaussian quadratic coefficients. The three population derivatives on $[-1, 1]$ are $x + 0.1x^2$, $0.15x + 0.05x^2$, and $\text{sign}(x)\{\max(|x| - 0.25, 0)\}^2$. The first two vanish uniquely at zero; the third vanishes on $[-0.25, 0.25]$. We use 400 participants, 1,000 replications, a 201-point grid, and 300 multiplier draws per replication.

Band coverage ranges from 0.944 to 0.952; inversion includes *every* true grid root with probability 0.968–0.969. The Wald calculation instead selects the estimated root nearest zero and assesses the designated root zero. Its coverage is 0.949 and 0.956 for the first two curves, but 0.243 under nonuniqueness (Figure 3). Thus the fixed nearly flat curve does not itself produce Wald undercoverage. A separate vanishing-curvature array in Appendix F exhibits

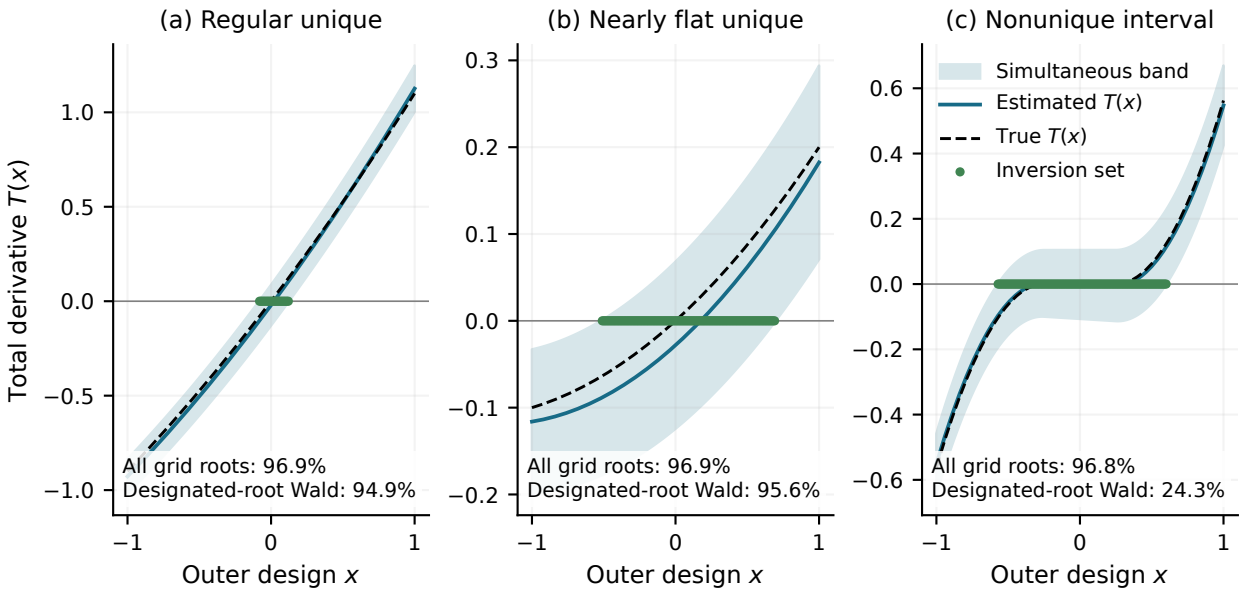


Figure 3: Outer geometry and band inversion. Curves show the first replication in each regime; green points mark the inverted set. Inversion coverage concerns all true grid roots simultaneously. Wald coverage concerns the designated root zero, a different event.

that failure as the local slope shrinks. These process experiments assess grid-based inference; the continuum theorem requires the uniform conditions in Section 4.4. They do not establish joint robustness to weak lower identification and flat outer geometry.

6 Applications to mobile health

HeartSteps and Drink Less illustrate how accounting for a learned response changes a benefit–burden comparison. We estimate proximal intervention benefit from randomized trial data and apply the smooth response in (11). The constructed intervention rules are evaluated over recorded contexts; they differ from the randomization rules that collected the data. HeartSteps provides the leading example, while Drink Less illustrates an unresolved design contrast.

6.1 Design criterion and uncertainty

Let $b(z) = z^\top \beta_b$ be proximal benefit, including an intercept, and $\pi_x(z) = \text{expit}\{[b(z) - x]/\tau\}$ the learned intervention probability. The penalty x has the same units as benefit. Using the augmented response in (12), we retain the benefit-maximization convention of Section 2.5:

$$\Phi(x) = \mathbb{E}_{\text{ref}}\{[b(Z) - x]\pi_x(Z)\} - \frac{\lambda}{2}(x - x_{\text{ref}})^2. \tag{30}$$

Here $\mathbb{E}_{\text{ref}}$ averages recorded reference contexts. The direct and response-adaptation derivatives are

$$\begin{aligned} D(x) &= -\mathbb{E}_{\text{ref}}\{\pi_x(Z)\} - \lambda(x - x_{\text{ref}}), \\ C(x) &= -\tau^{-1}\mathbb{E}_{\text{ref}}\{[b(Z) - x]\pi_x(Z)[1 - \pi_x(Z)]\}, \qquad T(x) = D(x) + C(x). \end{aligned} \tag{31}$$

We fix $(\tau, \lambda, x_{\text{ref}}) = (0.50, 0.70, 1.40)$ on $[0, 2]$ for HeartSteps and $(0.03, 12, 0.12)$ on $[0, 0.20]$ for Drink Less. These choices define distinct targets on the two benefit scales.

The selected decisions $x_D^\star, x_T^\star$ are the largest interior sign-changing crossings, with $\Delta_x = x_T^\star - x_D^\star$. The total crossing is a selected stationary point of (30). Positive Δ_x gives higher penalties and lower intervention probabilities at every fixed context.

Primary intervals use 5,000 paired resamples of centered participant coefficient influence contributions, recomputing both roots and their intensities in every draw. All draws have both roots. Features, preprocessing, and nuisance fits remain fixed, so these intervals describe coefficient uncertainty conditional on recorded contexts. The calculations use the analytic response rather than the numerical multi-adjoint implementation of Section 3. Separate

Table 3: Application stationary roots and recorded-opportunity intervention intensities.

Quantity	Direct root	Total root	Total minus direct
HeartSteps ($n = 37$)			
Stationary roots	1.261	1.526	0.265
95% interval	$[1.196, 1.298]$	$[1.513, 1.531]$	$[0.222, 0.323]$
Suggestions/protocol day	0.386	0.239	-0.148
95% interval	$[0.284, 0.568]$	$[0.187, 0.328]$	$[-0.242, -0.096]$
Drink Less ($n = 349$)			
Stationary roots	0.077	0.087	0.010
95% interval	$[0.066, 0.087]$	$[0.055, 0.102]$	$[-0.012, 0.016]$
Notifications/30 days	15.432	13.855	-1.577
95% interval	$[11.802, 19.266]$	$[9.856, 21.167]$	$[-2.489, 1.977]$

Intervals use 5,000 paired participant coefficient influence-contribution resamples, conditional on the frozen empirical features and preprocessing. Both roots and intensities are recomputed in each draw. HeartSteps standardizes 6,179 available records over 37×42 participant protocol days; Drink Less standardizes 30 records per participant. Roots follow the largest interior crossing rule. Intensities describe the observed record distribution, not a future history law.

HeartSteps simultaneous inverse sets use 2,000 participant resamples with model refitting and renewed feature standardization. Appendix G gives both procedures and their distinct targets.

6.2 HeartSteps

HeartSteps V1 (Klasnja et al., 2019) contributes 37 participants and 6,179 available decision times over 42 protocol days. The outcome is the public zero-imputed post-decision step count, transformed as $\log(\text{steps} + 0.5)$. Cross-fitted treatment-contrast pseudo-outcomes estimate benefit with pre-decision activity and study day as moderators. The fitted day interaction is -0.2348 per standardized day, with 95% interval $[-0.3582, -0.1114]$.

The selected direct and total penalties are 1.26115 and 1.52611. Their paired difference is 0.26496, with interval $[0.22196, 0.32338]$. Accounting for adaptation therefore resolves a shift toward a higher penalty under this specification. The standardized suggestion intensity

decreases by 0.148 per protocol day, with interval $[-0.242, -0.096]$ (Table 3). The separately constructed simultaneous inverse sets are connected and disjoint.

The small participant sample warrants a qualification. A related scalar calibration generator uses square-root steps and $\tau = 0.35$, rather than the full observed log-step fit with $\tau = 0.50$. At $n = 37$, its paired interval coverage is 0.948 among 998 valid intervals out of 1,000 replications, or 0.946 counting undefined intervals as misses. Coverage conditional on a connected total inverse set is 0.841, and mean width is 0.724, versus the observed 0.10142. These are generator-specific geometry and precision diagnostics, not a confidence level for the observed interval. The larger end-to-end study does not validate this application's interval; Appendix G.7 gives the full calibration results.

6.3 Drink Less

Drink Less (Bell et al., 2023) contributes 349 participants and 10,470 participant-days, with 30 days per participant. The proximal outcome is app opening during 8–9 PM, rather than alcohol consumption. Weighted and centered least squares on the risk-difference scale (Boruvka et al., 2018) uses study day, engagement before 8 PM, previous-day evening engagement, and the previous notification as moderators.

The selected direct and total penalties are 0.07713 and 0.08710. Their difference is 0.00996, with interval $[-0.01203, 0.01573]$. The notification-count difference is -1.577 per 30 recorded days, with interval $[-2.489, 1.977]$ (Table 3). Both contrasts remain unresolved; the fitted point direction alone does not establish a change in intervention intensity.

Sensitivity to the benefit definition also matters. A declining risk difference can reflect

changing baseline engagement as well as moderation of a relative effect. In a separate risk-difference sensitivity derived from an estimated marginal excursion effect on the log-relative-risk scale (Qian et al., 2021), fixing the explicit baseline-day term reverses the point direction of the decision contrast; its interval still includes zero. Appendix G.6 reports these separate specifications and the calibration and derivative-target details.

The reproducibility materials provide frozen inputs, programs, seeds, and figure-generation commands. Numerical calculations use NumPy and SciPy (Harris et al., 2020; Virtanen et al., 2020).

7 Discussion

Statistical inference for the total derivative of an outer criterion must account for the lower response induced by a design change. The decomposition into direct and chain components applies to the general response-equation framework; the chain component has a policy-mediated interpretation when the response parameterizes a policy. This decomposition separates uncertainty about a derivative component from uncertainty about a stationary decision. A nonzero chain component at one design value need not resolve a shift in a selected stationary value: its decision relevance also depends on where it occurs and on local curvature, as formalized by Proposition D.11.

The guarantees distinguish the source of weak information. For the chain component at a fixed design value, identification strength determines attainable finite-sample accuracy in the specified affine–quadratic class. The stabilized estimator uses specified class and identification bounds; adaptation to an unknown identification bound remains open. Moment

projection provides weak-identification coverage for that component under its standardized conditions; at exact singularity, the guarantee concerns each selected moment-representable value. Simultaneous stationary-set inference instead retains regular lower identification while allowing flat or nonunique outer roots. Extending these guarantees jointly, and obtaining matching finite-sample bounds for the full nonlinear trajectory model, are natural next problems.

The nonlinear simulation evaluates the complete numerical lower-response and adjoint pipeline. Operational tolerances yield near-nominal calibration on the specified design grid, while a looser lower solve can leave material simultaneous undercoverage despite substantially better pointwise behavior. Orthogonality controls leading sensitivity to nuisance error; it does not remove finite-sample numerical remainder. The controlled studies separately examine weak lower identification and nonregular outer geometry.

The applications illustrate proximal benefit–burden comparisons under prescribed response rules and recorded reference contexts. HeartSteps resolves a root difference under the fitted specification, with a geometry-specific calibration caveat and unusually narrow observed precision relative to its generator. Drink Less leaves the root contrast unresolved and illustrates sensitivity to the benefit specification (Appendix G). Temperatures, quadratic reference terms, and benefit specifications define these targets. The penalty is a price for intervention frequency, not an estimated dual multiplier from an explicitly solved constrained problem. The intensity intervals condition on recorded contexts and have not received a separate coverage study; future histories, availability, and distal health effects require additional analysis.

Known randomization and optional Markov restrictions can improve precision within their

respective statistical models. Richer response models, including policy classes, require control of identifiable directions, score envelopes, and nuisance errors across the design interval. These extensions preserve the central inferential questions: how the learned response changes the derivative, and how precisely that derivative determines an outer decision.

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